

Interacting By Drawing: Introducing Machine Learning Ideas to Children at a K–9 Science Fair

Tapani Toivonen
Ilkka Jormanainen

Matti Tedre
firstname.lastname@uef.fi
School of Computing, University of
Eastern Finland
Joensuu, Finland

Radu Marinescu-Istodor
radu.marinescu-istodor@karelia.fi
Karelia University of Applied Sciences
Joensuu, Finland

Teemu Valtonen
Henriikka Vartiainen
Juho Kahila
firstname.lastname@uef.fi
School of Applied Educational
Science and Teacher Education,
University of Eastern Finland
Joensuu, Finland

ABSTRACT

In the 2010s machine learning (ML) became a key driver of quickly growing number of apps and services. How to teach ML and other data-driven approaches in K–12 education has become a focus of intensive research efforts, with many recent advances in technology, pedagogy, and classroom integration. What to teach learners, at what age, and how, are some of the open questions being explored.

This study explored children’s interactions with a simple image classification tool, which used only two features to classify images. The results offer a proof-of-concept of how to teach 1) the principles of the ML workflow, and 2) some central ML insights, including image recognition, supervised learning, training data, model, feature, classifying, and accuracy. The results recognize how learning the principles of technology facilitates a shift in the locus of explanation from what oneself does to what the computer does. The results provide examples of how to support children’s developing data agency.

CCS CONCEPTS

• **Human-centered computing** → **User studies**; • **Social and professional topics** → *Computing education*; *K-12 education*; • **Computing methodologies** → *Machine learning*.

KEYWORDS

Machine learning; educational technology; artificial intelligence; K-12; school; computing education; 21st century skills

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1 INTRODUCTION

In parallel with the rapid rise of machine learning (ML) based technology, in the past five years the computing education field has seen a resurgence of efforts to bring artificial intelligence (AI) and ML education to younger and younger groups of learners [4, 9, 11, 19]. That shift has been driven by several developments: Firstly, it has been recognized that ever-younger children have become active users of services that use user data and ML for privacy-intrusive purposes; take, for instance, YouTube, Whatsapp, Spotify, Snapchat, and TikTok. Secondly, it has become obvious that understanding how ML-based, data-driven automation works requires skills and competences that traditional computational thinking as taught today cannot reach [2, 15, 18]. Thirdly, the past few years have seen a quickly growing number of initiatives for teaching AI/ML in K–12 (school) education: Among other things researchers have presented curricular extensions [3, 8, 16, 23], classroom material and activities [6, 14], MOOCs [4], ethics [1, 9] principled frameworks [9, 21], pedagogical outlines [1, 22] and educational technology [10, 17, 25]. Many AI/ML education projects in the K–12 context are driven by the quest to support the development of today’s youth to become informed, critical, and skilled actors in a datafied world dominated by services, applications, and systems built on constant data collection and ML [5, 9, 19]. A rich and deep picture of the state of AI/ML education in K–12 emerges from several surveys that have mapped different aspects of the field [9, 11, 19, 24].

The commodification of machine learning tools has brought “low-floor” [12] ML education technology to the K–12 classroom. ML is especially good at many application areas and kinds of data which traditional programming struggles with: unstructured media, video, sensor data, sound, and pictures. Programming a face recognition application is difficult, but creating a ML model for the same purpose is a matter of providing a ML algorithm many enough examples. There is a growing body of literature on children learning some principles of AI/ML through media-rich ML educational technology, such as speaker recognition, body pose recognition, speech synthesis, sports activities through sensors, gestures, and many others [7, 9, 11, 19, 24].

New easy to access educational technology for teaching ML/AI [11] has fueled research on “low-floor” interactive environments for ML/AI education in schools. Researchers have studied their classroom integration, their pedagogy, and other related educational aspects. But as the computing education research body of

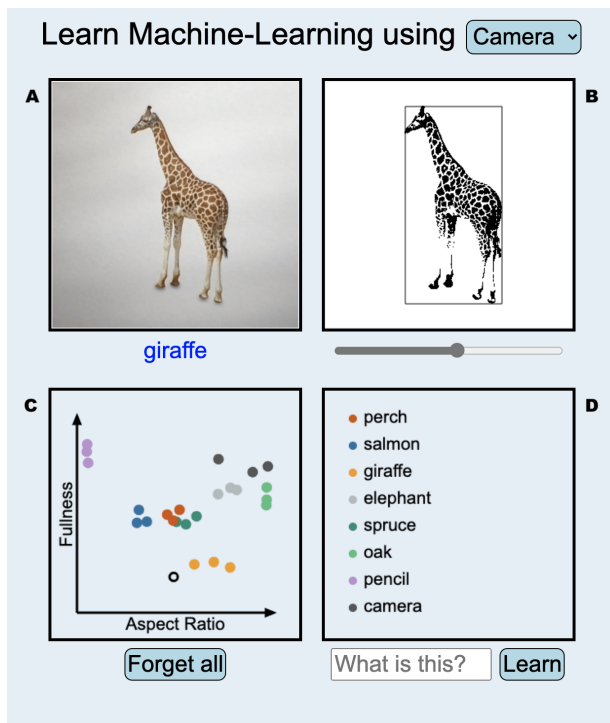


Figure 1: User interface of the image classification tool.

literature nearly completely pertains to classical programming approaches [15], and as ML and neural networks employ conceptually and technically very different principles, there is a need to expand computing education research and educational technology research efforts towards ML/AI [18]. This paper reports a case study of teaching basic machine learning concepts (image recognition, supervised learning, training data, model, feature, classifying, accuracy) to children aged from 5 and older using a simple image classifier tool developed for this purpose. The paper reports children's and their guardians' interactions with the tool, their workarounds and reasoning when the system did not behave as expected, and their emerging understanding of how a simple image classifier works.

2 METHODOLOGY AND WORKSHOP DESIGN

The intervention that was explored was a ML/AI workshop in SciFest, the largest children's science fair in Finland. The festival took place on September 4, 2021, in Joensuu, Finland, and the participants were a mixture of school groups and families. The festival featured a wide range of workshops, science cafés, competitions, and other types of program in all STEM fields (science, technology, engineering, and mathematics). The ML/AI workshop was visited by several hundreds of young visitors from the region, and they were aged 5 years and older. The workshop did not have specific time slots or reserved seats, but the participants were able to join the activities at any time. Of the visitors, roughly 150 children actively participated in the approximately 15 minutes long workshop, and many children were able to participate at the same time.

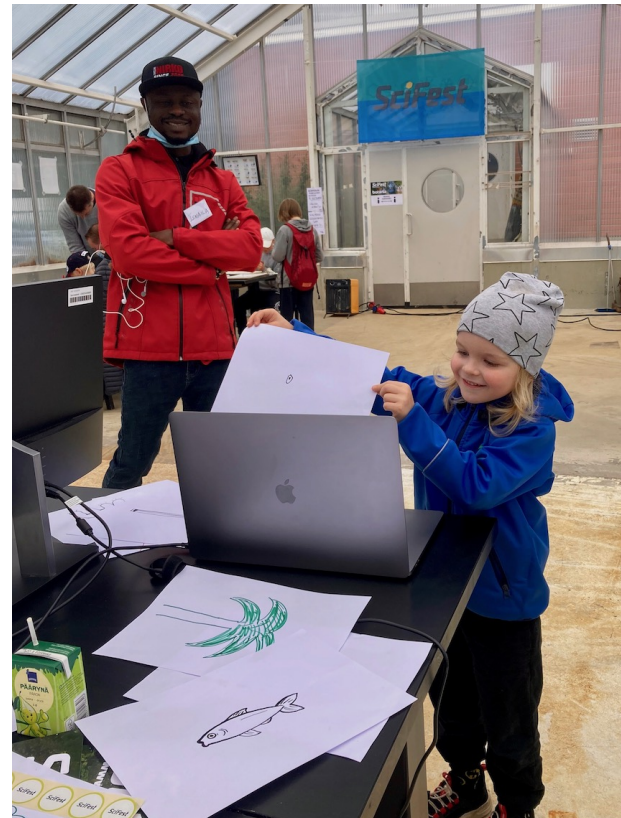


Figure 2: The ML/AI workshop at the SciFest science fair.

The theme of the workshop was image recognition. This study explored how children in that workshop interact with an image recognition system (classifier), using their own drawings as the objects to be classified. The workshop featured a simple learning application [10] that used the computer's webcam for capturing image data at live speed and that classified each frame using a simple feature-based classifier. The theme of classification was animals: Children were told that the system currently has been trained with a certain set of animals, and they can teach it more if they like. The interface is simple, consisting of four views: Fig. 1 (A) A small video screen of the webcam feed; Fig. 1 (B) The current frame as the model sees it (showing aspect ratio and fullness); Fig. 1 (C) All currently labeled images on a 2D feature space; and Fig. 1 (D) A complete list of labels. In addition, under view (A) a small label shows the classification result for the current webcam frame, and under view (D) a small text box and a "Learn" button that enables adding the current webcam image labeled by the text box. The workshop was equipped with computers and external monitors to showcase the machine learning tool for the visitors. In addition, the workshop had paper sheets and color pens available for the participants to draw their own pictures of the objects to be detected (Fig. 3).

The pedagogical design of the workshop activities was based on participatory learning that emphasizes children's active contribution in shared meaning-making and endeavors [13]. This highlights

the creation of learning environments that provide children with opportunities to explore ML in an interest-driven and inquiry-oriented manner [22]. In this workshop, the children had the opportunity to create their own material artefacts (drawings) and use their own artwork for exploring the conceptual ideas related to image recognition systems.

3 FINDINGS

What insights participants gained from the workshop depended, first and foremost, by the time they spent at the workshop. Most visits were variants of three types of experience:

- (1) Many visitors visited the workshop only very quickly and tried the image recognition system without waiting to get any explanation of how image recognition and the system works. That group often left frustrated for not getting their expected results (the computer misclassified their drawings), and often blamed their own drawing skills. ("Maybe I just can't draw.")
- (2) Many of those who stayed longer continued to explore the system on their own, and when faced with misclassification, focused on trying to improve their pictures with details that do not affect the model's classification results (e.g., many added details unique to each type of animal, such as drawing scales on the fish).
- (3) Many visitors who, perhaps after a failed attempt, stayed for a guided exploration of the system were explained how image recognition works, what are features (and what features did the current system use), and how the model "sees" the world.

The third group showed that it was possible to in just 15 minutes shift the locus of explanation from what oneself does to what the computer does. That was done in two steps. The workshop tutor first introduced learners to an oversimplified image recognition application that used one single feature—image aspect ratio, calculated through bounding box—to classify images. Children quickly found out that from the perspective of a model that uses just one feature, a long fish and a tall giraffe look the same.

Then, after extending the model by a second feature—fullness, calculated by the number of dark pixels in the picture—the workshop tutor showed how the model is able to classify a large number of animals surprisingly well. What is more, as the tool visualized both two features the model uses (aspect ratio and fullness), the third group of visitors typically seemed to have no trouble understanding how the image recognition system works. For instance, after finding out that a line drawing of an elephant has very low fullness (and therefore is easily misclassified) learners would improve their picture, coloring the elephant, to improve the classification accuracy. They would make too skinny elephants wider to change the bounding box size (and to make them more elephant-like). They would draw giraffes tall and skinny legged to affect the aspect ratio and fullness.

In this child-driven interaction, children engaged in the process of inquiry as they were iteratively improving their drawings and at the same time, developing a deeper understanding of how image recognition works. In this process, the children also had the opportunity to explain, interpret, and share their own observations with



Figure 3: A 10-year old learner drawing an animal for inclusion in the training data set.

their peers, parents, as well as with the computer science experts, who were also providing encouragement and metacognitive support for children on an on-demand basis. In essence, the workshop appeared to support children's sense of agency by providing them with tools for making and thinking as well as the opportunities to advance evolving ideas and explanations in a supportive social situation.

Most children in the group 3 (visitors who stayed for the full guided explanation) showed emerging understanding of the main concepts: the idea of image classification, that the app they used classified images by two simple features, that the model's goodness (accuracy) is based on the quality of the training data, that the classification needed somebody (in this case a human) to label the training data (supervised learning), and that (in addition to better data) additional features could improve the accuracy of the model.

4 DISCUSSION

This case study presents another proof-of-concept for designing simple tools for teaching some key ML/AI insights to young learners. Similar to other similar case studies (e.g., [1, 3, 6, 7, 10]), it shows that children have no trouble understanding some basic ML/AI concepts, the ML training workflow, how to improve the accuracy of the model (by better data or added features), and when does the model fail to classify correctly and why.

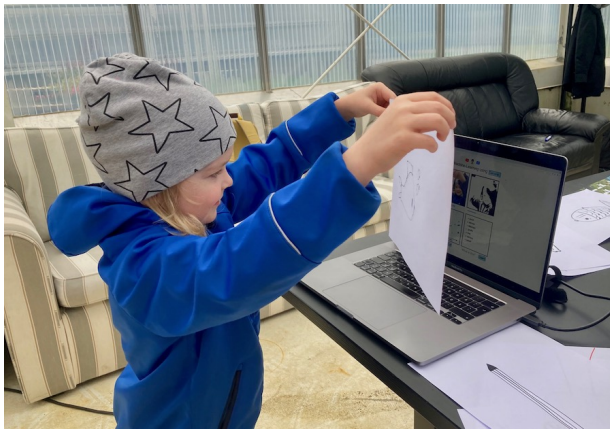


Figure 4: A six-year old learner testing the classifier with a new drawing of a fish not in the training set.

This work differs from the previous studies in one important respect: In this study children produced their own data and explored two features of the model. The results show how to present two key concepts of ML, model and feature, in a way that young children can readily understand (using vocabulary suitable for K–12).

One central lesson was the importance of shifting the locus of explanation from what oneself does to what the computer does. At the beginning almost all learners explained the failure of the model to classify images properly by their own drawing skills. Those learners who took the guided tutoring were after the lesson typically able to explain, in terms of the model’s features and limitations, why the model did not classify their drawing right.

A shift in locus of explanation is one important element of children’s developing sense of data agency (people’s volition and capacity for informed actions that make a difference in their digital world) [20]. These results suggest that by using very simple models (just a few features), it is possible to teach young learners how the model “sees” the world, as well as some of the important concepts related to representations, learning, and limitations of AI [21], as well as where and why “failure” happens.

5 CONCLUSIONS & WHAT’S NEXT

The popular approach to K–12 computing education, computational thinking (CT), is nearly entirely focused on traditional, rule-based programming. It struggles to explain to young learners how most of their digital tools and applications work. For example, it is incredibly difficult to program a rule-based classifier that can reliably tell apples from oranges. Children cannot learn why their phone can recognize their face by learning traditional programming. ML/AI based technology is common in children’s everyday lives, but learning how it works requires completely new approaches to computing education and educational technology. Those approaches and shifts in thinking in K–12 education are central to ML-enhanced computational thinking, “CT 2.0” [18].

There is a significant rise of interest in educational technology and designs for learning that can be used to teach children the

basics of ML/AI technology. Many initiatives have shown that data-driven approaches to computing can be introduced at earlier years than rule-driven approaches can [9, 11, 19, 24]. They do not require learning of complex syntax or coding, they enable bodily forms of interaction, and they shine in media-rich environments [19]. The early initiatives have suggested that already in the first grades of schooling children are able to learn some key insights related to creating ML models, such as image, sound, and pose classifiers. Research on how to teach AI in K–12 is, however, at a very early stage. This case study explored children’s interactions with a simple image classifier tool in a workshop that was aimed at introducing some basic ML/AI concepts: image recognition, supervised learning, training data, model, feature, classifying, and model accuracy.

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